

Technology Acceptance Model

GLOSSARY 04 - SHEET 7 / 8

Perceived usefulness and perceived ease of use predict intention to use a system

PROFESSIONAL DEFINITION

The Technology Acceptance Model holds that an individual's intention to use an information system is determined principally by two beliefs: perceived usefulness, the degree to which the person believes the system will enhance their job performance, and perceived ease of use, the degree to which they believe using it will be free of effort. Perceived ease of use also influences perceived usefulness, and usefulness is the stronger predictor of the two (Davis, 1989).

WHO DEFINED IT ORIGINALLY

Fred D. Davis developed the model in his 1985 MIT Sloan doctoral dissertation and published it in MIS Quarterly in 1989. It adapts Fishbein and Ajzen's (1975) theory of reasoned action to the technology context. Venkatesh and Davis (2000) extended it as TAM2, and Venkatesh, Morris, Davis and Davis (2003) consolidated eight competing models into UTAUT.

WHY THIS DEFINITION WAS DEVELOPED

Organisations were investing in systems that users declined to use, and the available explanations were post hoc. Davis sought a model that would predict acceptance from beliefs measurable before deployment, so that design and communication could be corrected while correction was still cheap.

HOW IT IS USED

Measure the two beliefs among the intended user population during design or pilot; a usefulness deficit requires a change to the system's fit to the job, an ease-of-use deficit requires a change to interface or training, and the two are not interchangeable.

WHEN IT IS USED

During design of an AI decision-support tool, at pilot before scaling, when usage is below plan after go-live, and when comparing two candidate vendor products for the same task.

BY WHOM IT IS USED

Product owners and user-experience designers, adoption and change leads, procurement teams evaluating vendor systems, and business sponsors deciding whether to scale a pilot.

FIGURE 1 - TWO BELIEFS, ONE BEHAVIOUR

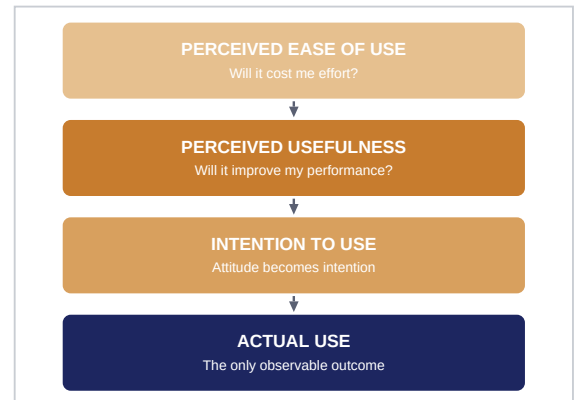


Figure 1. Ease of use acts on behaviour partly through usefulness, which is the stronger predictor. A system that is effortless and not useful is not adopted; the reverse is frequently adopted despite complaint (after Davis, 1989).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Its predecessor, the theory of reasoned action, was a general model of behaviour requiring a fresh belief-elicitation study for each context, which made it impractical for system evaluation. Davis fixed the two belief constructs in advance and supplied validated scales, so acceptance could be measured comparably across systems and organisations. UTAUT later added social influence and facilitating conditions, which matter particularly where use is mandated rather than voluntary - the usual case in enterprise AI.

RELATED DEFINITIONS IN THIS COURSE

[AI Adoption - Unit 4](#)
[Algorithm Aversion - Unit 2](#)
[Change Management - Unit 4](#)
[Technology Adoption Lifecycle - Unit 1](#)
[AI Literacy - Unit 2](#)

SOURCES (HARVARD REFERENCING STYLE)

Davis, F.D. (1989) 'Perceived usefulness, perceived ease of use, and user acceptance of information technology', MIS Quarterly, 13(3), pp. 319-340.
 Fishbein, M. and Ajzen, I. (1975) Belief, Attitude, Intention and Behavior. Reading, MA: Addison-Wesley.
 Venkatesh, V. and Davis, F.D. (2000) 'A theoretical extension of the technology acceptance model', Management Science, 46(2), pp. 186-204.
 Venkatesh, V., Morris, M.G., Davis, G.B. and Davis, F.D. (2003) 'User acceptance of information technology: toward a unified view', MIS Quarterly, 27(3), pp. 425-478.